

Chapter 1: Technology [8 Hours]

- **1.1 Definition**

1. Define technology. [2] (2078 Chaitra, Q1)
2. Define technology. [2] (2074 Magh, Q1)
3. Define technology. [Part of 8 (Probably 2)] (2079 Chaitra, Q1)
4. What do you mean by technology? [2] (2072 Ashwin, Q1)
5. Define technology. Discuss about the characteristics of technology with suitable example. [2+6] (2080 Chaitra, Q1)
6. Define technology is the foundation of civilization. [Part of 8 (Probably 4)] (2072 Magh, Q1)

- **1.2 Impact of Technology on Environment & Society**

1. Define technology and its impact on societies. [Part of 8 (Probably 4)] (2071 Bhadra, Q1)
2. Discuss the impact of technology on environment and society. [6] (2078 Chaitra, Q1)
3. How technological advancement places impact on environment and society? [Part of 8 (Probably 4)] (2073 Bhadra, Q1)
4. Write down benefit of technology on environment and society. [Part of 8 (Probably 6)] (2079 Chaitra, Q1)
5. Discuss with suitable examples on how development activities have impacts on environments and society. [8] (2079 Chaitra, Q4)
6. Discuss the impact of technology on society. [Part of 8 (Probably 4)] (2072 Magh, Q1)
7. Write the positive as well as negative impact of technology on society. [4] (2070 Bhadra, Q1)
8. How Nepalese society is impacted by technology in course of time? [Part of 8 (Probably 4)] (2069 Bhadra (Old Syllabus), Q4)

- **1.3 Benefits of Technology Due to New Inventions**

1. Discuss the benefits of modern technology for the promotion of innovative researches and their output application in various fields. [4+4] (2079 Jestha, Q1)
2. Write down benefits of technology in education sector. [6] (2072 Ashwin, Q1)
3. As a civil engineer how have you been benefited by technologies? [6] (2074 Magh, Q1)

- **1.4 Conflict of Technology, Technological Creates Opportunity for Society to Change**

1. Describe how technology helps create opportunity while transforming a society. [6] (2077 Chaitra, Q1)

- **1.5 Appropriate Technology**

1. Define appropriate technology with suitable examples. [2] (2077 Chaitra, Q1)
2. What is appropriate technology? [Part of 8 (Probably 1)] (2073 Bhadra, Q1)
3. Describe the role of appropriate technology on transforming a society. [8] (2079 Shrawan, Q1)
4. Define appropriate technology and its relevancy to LDC. [Part of 8 (Probably 4)] (2071 Bhadra, Q1)
5. What are appropriate technologies? How are they useful in least developed countries like Nepal? Justify with suitable examples. [8] (2075 Bhadra, Q2)
6. What are the elements for the sustainable development? How can you say Indigenous technology are sustainable in case of Nepal. [8] (2074 Bhadra, Q1)
7. What are the elements for the sustainable development? How can you say Indigenous technology are sustainable in case of Nepal. [8] (2073 Magh, Q1)
8. Write notes on: Appropriate Technology. [4] (2071 Magh, Q6b)

- **1.6 Intermediate Technology, Labor Based and Labor Intensive Technology**

1. Define Labor based and labor intensive technology with suitable example. [4] (2078 Kartik, Q1)
2. Discuss advantages and disadvantages of labor based and labor intensive development approach with suitable examples. [8] (2079 Chaitra, Q2)

- **1.7 Shifts in Employment Due to Technological Advancement**

1. Explain the technology provides shift in employment. [8] (2073 Bhadra, Q5)
2. Write short notes on: Technology and shift in employment. [4] (2069 Bhadra (Old Syllabus), Q6b)

- **1.8 Role of Technology to Unmask Old Social Problems, Society's Control of Technology**

1. How technology unmask old social problem? Justify your answer with suitable examples. [8] (2076 Bhadra, Q1)

2. How technology can control the price of basic commodities? Explain. [8] (2075 Bhadra, Q5)
3. "Technology can control price" Do you agree with this statement. Support your views with examples. [8] (2069 Bhadra (Old Syllabus), Q3)

- **1.9 Impact of Technology on Culture, Tradition and Societal Values**

➤ *No questions directly addressing this subtopic in the provided papers.*

- **1.10 Technology Is Irreversible**

1. Justify technology is irreversible. [4] (2078 Kartik, Q1)

- **1.11 Agricultural Age, Industrial Age and Information Age**

1. What is "Agricultural Age", "Industrial Age" and "Information Age"? Explain. [4] (2070 Bhadra, Q1)
2. What do you mean by an Information Society? Differentiate it with agricultural and industrial society. [8] (2076 Bhadra, Q3)
3. Define industrial revolution, industrial society and information society. [Part of 8 (Probably 4)] (2069 Bhadra (Old Syllabus), Q1)

- **1.12 Characteristics of Information Society**

1. What do you understand by information society? [8] (2072 Magh, Q2)
2. Explain information society. Also write down characteristics of information age. [7] (2071 Magh, Q1)
3. Describe briefly the characteristics of information society. [Part of 8 (Probably 3)] (2073 Bhadra, Q1)
4. What is meant by industrial and information society? Discuss the impacts of computer and cybernetics in creating information society that makes the whole world as a global village. [8] (2070 Magh, Q1)

- **1.13 Information as Power and Wealth**

➤ *No questions directly addressing this subtopic in the provided papers.*

Chapter 2: Development Approach [6 Hours]

- **2.1 LEP (Labor Based, Environment Friendly and Participatory)**

1. What is LEP? [Part of 7 (Probably 1)] (2071 Magh, Q2)
2. Describe briefly about the LEP development approach. [4] (2080 Chaitra, Q2)
3. Discuss LEP approach in infrastructure development. [4] (2071 Bhadra, Q3a)
4. Describe the LEP Approach for the development of Nepal? [Part of 8 (Probably 4)] (2072 Ashwin, Q2)
5. Discuss about the LEP development approach with both merits and demerits. [8] (2077 Chaitra, Q2)
6. How do you describe the LEP Approach for the development of Nepal? Explain advantages and disadvantages of LEP Approach. [8] (2074 Bhadra, Q5)
7. How do you describe the LEP Approach for the development of Nepal? Explain advantages and disadvantages of LEP Approach. [8] (2073 Magh, Q5)
8. What are the essential features of Participatory approach? Write down the advantages and disadvantages of public participation on developing project. [2+6] (2076 Bhadra, Q2)
9. Why participatory planning approach is important for sustainable infrastructure development? [4] (2079 Jestha, Q2)

- **2.2 Community Management, Engineers Role as Facilitator**

1. Define community management. [2] (2078 Chaitra, Q2)
2. Discuss about participatory approach for community management. [Part of 8] (2075 Bhadra, Q1)
3. Discuss about participatory approach for community management. [Part of 7 (Probably 3)] (2071 Magh, Q2)
4. Discuss about participatory approach for community management and engineer roles as facilitator in development activities. [4+4] (2079 Shrawan, Q2)
5. Discuss about participatory approach for community management and engineer roles as facilitator in development activities. [8] (2078 Kartik, Q2)

6. Discuss the role of engineers as facilitators in community-led development project. [4] (2080 Chaitra, Q2)
7. Discuss the role of engineer in community development. [8] (2073 Bhadra, Q4)
8. Discuss the role of Engineer in Community Management. [4] (2071 Bhadra, Q3b)

- **2.3 Key Features of Infrastructure Development Policies of Nepal**

1. Write some key features of infrastructure development policies of Nepal. [6] (2070 Bhadra, Q1)
2. What are the different key features of infrastructure development policies of Nepal? [3] (2070 Magh, Q3)

- **2.4 Ethnographic Approach to Collect Information**

1. What do you understand by 'ethnographic data collection approach'? [3] (2079 Jestha, Q2)
2. Discuss about ethnographic approach for community management. [Part of 7 (Probably 3)] (2071 Magh, Q2)

- **2.5 Participatory Approach as Community Empowerment**

1. Explain community empowerment process of LEP Approach. [Part of 8 (Probably 4)] (2072 Ashwin, Q2)
2. Describe how participatory approach encourages to empowerment community in development activity. [6] (2078 Chaitra, Q2)
3. How participatory approach encourage to empowerment community in development activity. [8] (2074 Magh, Q2)
4. How participatory approach encourage to empowerment community in development activity? [2] (2070 Bhadra, Q1)
5. Do participatory approaches empower community, justify your answer. [5] (2070 Magh, Q3)

- **2.6 Participatory Tools, Focus Group Discussions, Key Informants Interview**

➤ *No questions directly addressing this subtopic in the provided papers.*

- **2.7 Participatory Observation, Structured Questionnaire**

➤ *No questions directly addressing this subtopic in the provided papers.*

- **2.8 Resource Mapping, Wealth Ranking, Poverty Definition**

1. What do you mean by Resource Mapping? [Part of 8] (2075 Bhadra, Q1)

Chapter 3: Brief History of Human Civilization [4 Hours]

- **3.1 Early Civilization**

1. State the different ages of human development. [3] (2070 Magh, Q2)

- **3.2 Great Renaissance of Europe**

1. Describe in brief about the "Great Renaissance of Europe" with key features. [4] (2077 Chaitra, Q3a)
2. Discuss intellectual revival of Europe in 13 to 15 century. [5] (2071 Bhadra, Q7)

- **3.3 Early Part of Industrial Revolution**

1. Define industrial revolution. [2] (2079 Shrawan, Q3)
2. Describe briefly the impact of early part of industrial revolution. [3] (2070 Bhadra, Q1)
3. Define industrial revolution. Describe the environmental consequences industrial revolution. [1+3] (2080 Chaitra, Q4a)
4. Describe the major changes the industrial Revolution brought to human civilization. [8] (2073 Magh, Q4)

- **3.4 Transformation of Industrial Society into Information Society**

1. Discuss in brief on transformation of industrial society to information society. [4] (2078 Chaitra, Q3a)
2. Describe in brief an how industrial society transformed to information society. [4] (2078 Kartik, Q3)
3. Discuss about the transformation from industrial to information society. [Part of 8 (Probably 4)] (2069 Bhadra (Old Syllabus), Q1)
4. How industrial societies enter into information society? [5] (2071 Magh, Q3)

- **3.5 Impact of World War 1 & 2**

1. Discuss about world war I and world war II along with the positive and negative impacts. [8] (2074 Magh, Q3)
2. Describe the impact of World War I and II in our society. [5] (2070 Bhadra, Q1)
3. Write the key developments on technology during first and Second World War and their positive and negative impact on population. [5] (2070 Magh, Q2)

- **3.6 Rise of Environmental Issues**

➤ *No questions directly addressing this subtopic in the provided papers.*

- **3.7 Climate Change as a Threat to Human Civilization**

1. Justify with suitable examples on how climate change becomes threat to human civilization. [4] (2079 Chaitra, Q3a)
2. "Climate change is a Threat to Human Civilization!!". Justify it. [5] (2079 Jestha, Q3)
3. Justify, that climate a change has become a great threat to human civilization. [6] (2079 Shrawan, Q3)

Chapter 4: Environment [3 Hours]

- **4.1 Definition**

1. Define environment and ecosystem? [Part of 8 (Probably 2)] (2073 Magh, Q3)
2. Define environment, and ecosystem. [Part of 8 (Probably 2)] (2070 Magh, Q4)

- **4.2 Importance, Ecology & Ecosystem**

1. Define ecosystem and ecology. How increasing population disturbs to natural ecosystem and how to minimize the disturbance? [2+3] (2070 Bhadra, Q1)
2. What are the primary reasons for human interventions in the ecosystem? [8] (2074 Bhadra, Q3)
3. What are the primary reasons for human interventions in the ecosystem? How natural resources of ecosystem is converted into wealth? [8] (2072 Ashwin, Q3)

4. Write notes on: Ecology and Ecosystem. [4] (2071 Magh, Q6c)
5. Write short notes on: Ecology, ecosystem and its characteristics. [4] (2069 Bhadra (Old Syllabus), Q6a)

- **4.3 Conservation of Environment**

➤ *No questions directly addressing this subtopic in the provided papers.*

- **4.4 Optimum Utilization of Natural Resources**

1. What do you understand by-optimum utilization of natural resources for environment conservation? [3] (2079 Jestha, Q4)

- **4.5 Renewable and Non-Renewable Resources**

1. What are the renewable and non-renewable energy resources? [3] (2079 Jestha, Q4)
2. Differentiate between renewable and non-renewable energy. [5] (2071 Bhadra, Q6)

- **4.6 Conflict of Resources**

1. Describe about conflict of resources with suitable examples. [4] (2078 Chaitra, Q3b)
2. Discuss about conflict of resources with suitable example. [4] (2078 Kartik, Q3)

- **4.7 Global Environmental Issues**

1. Define regional and global environmental issues. And also discuss briefly about the factors contributing to global warming. [4] (2079 Chaitra, Q3b)
2. Discuss briefly on the Global Environmental issues and its driving force. [4] (2077 Chaitra, Q3b)

- **4.8 Environmental Issues of Nepal**

1. List out the major environmental issues of Nepal. [3] (2070 Bhadra, Q1)
2. Discuss about the major environmental issues of Nepal. [Part of 8 (Probably 4)] (2074 Bhadra, Q2)
3. List out the key environmental issue people facing in Kathmandu valley and discuss about the pressing factors of those environmental issue. [1+3] (2080 Chaitra, Q4b)

4. What are the major environmental issues in Nepal? Discuss along with the causes and mitigation measures. [8] (2074 Magh, Q4)
 5. Discuss the major environmental issues of Nepal? What are the causes of acid rain? [8] (2072 Ashwin, Q5)
 6. What is the major environmental problem of Nepal? [4] (2069 Bhadra (Old Syllabus), Q4)
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Chapter 5: Water and Air Pollution [6 Hours]

- **5.1 Fecal-Oral Infection Transmission Route**

1. Describe Fecal-oral infection transmission route with examples of primary and secondary barriers. [6+2] (2080 Chaitra, Q5)
2. Explain fecal-oral group of infections. [4] (2076 Bhadra, Q5)
3. Describe the fecal oral transmission route. [Part of 8 (Probably 4)] (2072 Magh, Q3)

- **5.2 Preventive Measures**

1. Discuss preventive measures to transmit fecal-oral group of infections. [4] (2076 Bhadra, Q5)

- **5.3 On-Site Sanitation (Including Eco-Sanitation)**

➤ *No questions directly addressing this subtopic in the provided papers.*

- **5.4 Importance of Health Education**

1. Discuss about the importance of health education. [Part of 8 (Probably 2)] (2073 Bhadra, Q2)
2. Discuss the role of the health education regarding the water pollution in environment. [Part of 8 (Probably 4)] (2072 Magh, Q3)

- **5.5 Organic Pollution**

1. What is organic pollution in water? Discuss in detail aerobic digestion and anaerobic digestion with practical examples. [8] (2073 Bhadra, Q6)

- **5.6 Inorganic Pollution (Nitrate, Fluoride, Iron, Manganese, Calcium Arsenic, Heavy Metal, Water Pollution Due to Insecticides and Pesticides)**

1. Describe how the disposal of sludge and industrial waste results in pollution of river, lakes and canals? Recommended appropriate mitigation measures to minimize the impacts. [Part of 8 (Probably 6)] (2073 Magh, Q3)
2. Describe how the disposal of sludge and industrial waste results in pollution of river, lakes and canals? Recommended appropriate mitigation measures to minimize the impacts. [Part of 8 (Probably 6)] (2070 Magh, Q4)
3. Write notes on: Pesticide Hazards in Nepal. [4] (2071 Magh, Q6a)
4. Describe about the industrial wastewater and its treatment methods. And, list out the major causes of polluting Bagmati River in Kathmandu. [8] (2069 Bhadra (Old Syllabus), Q2)

• **5.7 Sources, Causes & Impacts of Air Pollution**

1. What are the sources, causes and impact of air pollution? [5] (2079 Jestha, Q5)
2. What are the impacts of air pollution? [Part of 8 (probably 4)] (2079 Shrawan, Q4)
3. What are the sources, causes and impacts of air pollution? [4] (2078 Kartik, Q4)
4. What are the sources, causes and impacts of air pollution? [Part of 8 (Probably 6)] (2073 Bhadra, Q2)
5. What is air pollution? What are the causes and impacts of air pollution? [Part of 8 (Probably 6)] (2075 Bhadra, Q3)
6. Write down causes and effects of air pollution. [Part of 8 (Probably 6)] (2072 Magh, Q5)
7. Write down in detail causes and effects of air pollution. [Part of 8 (Probably 6)] (2071 Bhadra, Q2)
8. What are the sources of Air and Water Pollution? [4] (2070 Bhadra, Q4 OR)
9. List and the key causes of water and air pollution in the Kathmandu Valley. [4] (2078 Chaitra, Q4)
10. Explain the reason of air and water pollution in the Kathmandu valley with its major mitigation measures. [4+4] (2077 Chaitra, Q4)
11. Write short notes on: Air pollution, pollutant, its impacts and control measures. [4] (2069 Bhadra (Old Syllabus), Q6c)
12. Write short notes on: Acid rain, its impacts and control measures. [4] (2069 Bhadra (Old Syllabus), Q6d)

- **5.8 Mitigation Measures**

1. How can air pollution be mitigated? [Part of 8 (Probably 4)] (2079 Shrawan, Q4)
2. What are the mitigation measures of air pollution? [Part of 8 (Probably 2)] (2075 Bhadra)
3. Write down mitigation measures of air pollution. [Part of 8 (Probably 2)] (2072 Magh, Q5)
4. Write down in detail mitigation measures of air pollution. [Part of 8 (Probably 2)] (2071 Bhadra, Q2)
5. Recommend key mitigation measures for air pollution in Kathmandu valley. [4] (2078 Kartik, Q4)
6. How can we reduce air pollution of Kathmandu? [Part of 8 (Probably 4)] (2074 Bhadra, Q2)
7. Recommend mitigation measures to air and water pollution. [4] (2078 Chaitra, Q4)
8. Write some mitigation measures of water and air pollution. [4] (2070 Bhadra, Q4 OR)

- **5.9 Indoor Air Pollution**

1. What causes indoor air pollution? What could be done to reduce indoor air pollution in rural Nepal? [7] (2071 Magh, Q4)

- **5.10 Severity of Its Problems in Nepal**

1. Highlight the Nepal's effort to address environment protection issues. [4] (2079 Jestha, Q5)

Chapter 6: Climate Change [3 Hours]

- **6.1 Definition, Causes, Impacts**

1. What is climate change? [Part of 8 (Probably 4)] (2072 Magh, Q4)
2. Define climate change & climate variability. [2] (2080 Chaitra, Q3)
3. Define climate change with suitable example of extreme events. [4] (2078 Chaitra, Q5)
4. What are the factors contributing to the climate change? [Part of 8 (Probably 4)] (2075 Bhadra, Q4)
5. Describe causes and Impacts of climate change. [Part of 8 (Probably 4)] (2074 Bhadra, Q4)
6. List out the climate change gases. Explain the effect of climate change. [4+Part of 4 (Probably 2)] (2076 Bhadra, Q4)

7. Write short notes on: Causes and impacts of climate change. [4] (2078 Kartik, Q5a)
8. What are the factors contributing to climate change? Discuss about the impacts of climate change in case of Nepal. [8] (2074 Magh, Q5)
9. What do you mean by climate change? Write major symptoms of climate changes in Nepal. [Part of 8 (Probably 6)] (2073 Magh, Q2)
10. List out the challenges of climate changes being faced by our country. [2] (2080 Chaitra, Q3)
11. What are the factors contributing to global warming? Outline their sources, impacts and mitigation measures of global warming. [8] (2072 Ashwin, Q4)
12. What do you mean by climate change? Write major symptoms of climate changes in Nepal. [1+4] (2070 Magh, Q5)
13. What is climate change? Describe its impacts and potential mitigation measures. [1+4] (2070 Bhadra, Q5)
14. What is climate change? Write down its probable impacts and problems mitigation measures that Nepal can adopt. [5] (2071 Bhadra, Q5)
15. What is climate change? Describe its impacts and potential mitigation measures. [Part of 8 (Probably 6)] (2069 Bhadra (Old Syllabus), Q5)

- **6.2 Mitigation Measures**

1. What would be the appropriate mitigation measures to reduce the effect of climate change? [Part of 8 (Probably 4)] (2072 Magh, Q4)
2. Discuss about climate change and adaptation measure from Nepalese perspective. [8] (2073 Bhadra, Q3)
3. Write short notes on: Mitigation measures to climate change. [4] (2078 Kartik, Q5c)
4. Describe its impacts and potential mitigation measures. [2] (2069 Bhadra, Q5)

- **6.3 International Efforts to Mitigate Its Problems**

1. Write down international efforts. [Part of 4 (Probably 2)] (2076 Bhadra, Q4)
2. State some international efforts for mitigation and adaptation of climate change induced problems. [Part of 8 (Probably 2)] (2073 Magh, Q2)
3. Elaborate the International efforts to combat climate change effects. [5] (2079 Jestha, Q6)

4. Describe about key international efforts to mitigate climate change impacts. [4] (2078 Chaitra, Q5)
5. Explain briefly the international efforts towards mitigation measures undertaken globally. [Part of 8 (Probably 4)] (2075 Bhadra, Q4)
6. Describe international efforts made to address climate change. [3] (2070 Bhadra, Q5)
7. State some national and international efforts for mitigation and adaption of climate induced problems. [3] (2070 Magh, Q5)
8. Discuss about national and international efforts to tackle climate change. [4] (2080 Chaitra, Q3)
9. What are the key international efforts to combat climate change effects? How Nepal can be streamlined to it? [8] (2079 Chaitra, Q5)
10. Write short notes on: Climate Change and International Efforts. [4] (2077 Chaitra, Q5b)
11. Describe international efforts made to address this problem. [Part of 8 (Probably 2)] (2069 Bhadra (Old Syllabus), Q5)

- **6.4 Bio-Gas, Organic Farming**

1. Discuss on, how Bio-gas and organic farming help minimize climate change. [8] (2079 Shrawan, Q5)
2. Write short notes on: Bio-gas and its suitability in Terai Region of Nepal. [4] (2077 Chaitra, Q5a)

- **6.5 Deforestation and Its Consequences**

1. Write short notes on: Deforestation and its consequences. [4] (2077 Chaitra, Q5c)

- **6.6 Importance of National Parks, Conservation Areas and Forestation in Nepal**

1. Describe the importance of national parks, conservation areas and forestation programs in Nepal as an effective strategy to reduce green house gas emission. [6] (2071 Magh, Q5)
2. How can we protect our National Parks and Conservation Areas. [Part of 8 (Probably 4)] (2074 Bhadra, Q4)
3. Write short notes on: National Park and its importance. [4] (2078 Kartik, Q5b)